IAMS AI Guideline and Compliance

AI-Task Force on Molecular Sciences (ATOMS)
Institute of Atomic and Molecular Sciences (IAMS)
Academia Sinica, Taiwan
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1. Purpose and Scope

Purpose: This guideline clarifies how existing standards for responsible research and administrative practice apply to the use of artificial intelligence (AI) tools at IAMS. AI tools are treated as assistive tools that do not alter the responsibilities of researchers or staff. It aligns with Taiwan's AI Basic Act (effective January 2026), Academia Sinica's Guidelines for Using Generative AI Tools (February 2026), and international best practices for AI in academic research.

Scope: This guideline applies to all IAMS personnel — including research fellows, postdoctoral researchers, graduate students, research assistants and administrative staff — in both research and administrative activities.

2. Regulatory Framework

Taiwan AI Basic Act (2026): Establishes seven fundamental principles — sustainable development, human autonomy, privacy protection, cybersecurity, transparency, fairness, and accountability. Requires government entities to conduct risk assessments and establish internal AI use rules within 12 months.[1]

Academia Sinica Generative AI Guidelines: Three core principles — accuracy of AI-generated content, privacy and information security, and research transparency and accountability.[2]

International Standards: Alignment with COPE (Committee on Publication Ethics), ALLEA European Code of Conduct for Research Integrity, and publisher-specific AI policies (Nature, Science, Elsevier, Wiley, IEEE).[3,4]

3. Guiding Principles

The use of most AI tools does not fundamentally alter existing responsibilities, but may introduce additional considerations related to data handling, reproducibility, and system behavior. Activities remain governed by existing standards of research integrity, data protection, and professional accountability. These principles align with Academia Sinica's core requirements of accuracy, privacy protection, and transparency. Where specific considerations arise from AI use (e.g., data handling, reproducibility, or system behavior), these are addressed within the existing framework of research integrity and professional responsibility.

A. Responsibility and Research Integrity

- Researchers and staff bear full responsibility for all outputs, whether produced with or without AI assistance.
- All outputs must be verified to ensure accuracy of information, data, and citations.

- AI systems shall not be listed as authors on any publication, as authorship requires accountability that AI cannot fulfill.
- AI use must comply with established research and administrative ethics standards. Fabrication, falsification, or plagiarism, including through AI tools, constitutes research misconduct.
- Where AI tools are used in data analysis or scientific workflows, sufficient documentation should be maintained to enable independent verification of results.
- Researchers must verify AI-generated citations, references, data, and factual statements prior to use.

B. Data Protection and Information Security

- Existing requirements for data protection, confidentiality, and information security apply equally to the use of AI tools, including compliance with Taiwan's Personal Data Protection Act (PDPA).
- Confidential, sensitive, or personal data must not be input into AI systems that may retain, share, or use such data for model training or external use. This includes unpublished manuscripts, raw experimental data, peer review materials, proprietary information, and personal data unless explicitly permitted by institutional policy.
- Use of AI tools with research data is permissible where appropriate safeguards are in place (e.g., enterprise systems, API-based tools, or local deployments that do not use input data for training or external sharing), and where such use does not violate confidentiality obligations or institutional policies.
- AI tools shall not be used in contexts involving confidentiality obligations (e.g., peer review, proposal evaluation), unless explicitly permitted under applicable policies.
- Where available, institutionally approved or enterprise AI tools with appropriate data protection safeguards should be preferred.

C. Transparency and Disclosure

- Where AI tools are used substantially in research or administrative processes, such use shall be appropriately documented and disclosed in accordance with institutional, funder, and publisher requirements. Substantial use typically includes AI-assisted drafting of manuscripts, data analysis or interpretation, generation of figures, or other contributions that materially affect outputs.
- Users should retain sufficient records of significant AI use (e.g., tools used and general purpose) to support transparency, reproducibility, and compliance with applicable requirements.

D. Fairness and Non-Discrimination

- AI tools shall be used in ways that avoid bias, discrimination, or inequitable outcomes.

E. Risk Awareness

- For uses involving sensitive data, external AI systems, or autonomous operation, users should assess risks related to data protection, security, and reliability, and adopt appropriate safeguards where necessary.

4. Operational Guidance on Data Protection and Information Security

This section provides additional operational clarification of Section 3B and does not introduce new requirements. The use of AI tools is subject to existing institutional policies on data protection, confidentiality, and information security, including the Personal Data Protection Act (PDPA).

- Do not input classified or confidential data (e.g., personal data, aggregated financial data, or proprietary data) into external AI platforms.
- Do not input data into AI systems that allow such data to be retained, shared, or used for training. Where appropriate, use institutionally approved AI tools that provide suitable data protection guarantees. Users shall be aware of and assess the data handling and privacy policies of AI service providers before use.
- For higher-risk technical implementations, existing institutional IT and security policies apply.

5. Autonomous and Agentic Systems

Autonomous and agentic AI systems can pose challenges in attributing accountability and responsibility. Researchers and staff who initiate or operate agentic or autonomous AI systems retain ultimate responsibility for the actions of these systems. Such systems shall be used with appropriate safeguards and monitoring to prevent unintended or uncontrolled outcomes. Researchers and staff who are concerned about the potential for unexpected outcomes from operating autonomous or agentic systems are encouraged to discuss their system with the ATOMS Task Force.

6. Examples of Permitted and Prohibited Uses

AI tools may be used as assistive tools across research and administrative activities, subject to the general principles in Section 3.

Researchers and staff retain full responsibility for all outputs.

The following examples are illustrative and non-exhaustive.

6.1 Examples of Permitted Activities

- Drafting routine communications.
- Document formatting and template creation.
- Meeting summary drafting
- Translation assistance (users should exercise discretion to avoid disclosure of sensitive information and verify accuracy).
- Literature search and summarization.
- Language editing and grammar polishing.
- Software development, debugging and optimization.
- Data visualization formatting.
- Brainstorming and idea generation.
- Tutorial generation and learning/educational assistance.
- Generation of internal documentation.
- Manuscript drafting assistance (researcher retains full responsibility; disclosure required where use is substantial and in accordance with journal and funding provider policies).
- Statistical analysis and data interpretation support (researcher retains full responsibility; sufficient documentation should be maintained to enable independent verification).
- Figure and illustration generation (researcher retains full responsibility; disclosure and verification required in accordance with journal and funding provider policies).

- Grant proposal writing assistance (researcher retains full responsibility; disclosure required where applicable under funding provider policies).

6.2 Examples of Prohibited Activities (activities violating section 3 guidelines)

- Inputting classified or confidential institutional data into external AI systems.
- Automated decision-making on personnel matters without human oversight.
- Using AI to generate official institutional statements or communications without human review and approval.
- Use of AI tools in confidential processes (e.g., peer review, proposal evaluation) where disclosure is not permitted.
- Any use of AI that violates applicable laws, institutional policies, or research and publication standards.

7. Compliance and Implementation

- The IAMS Director has designated the **AI-Task Force for Molecular Sciences (ATOMS, IAMS, AS**) to oversee implementation.
- This guideline will be periodically reviewed and updated as needed.
- IAMS shall provide training and orientation on responsible AI use for all new personnel (**via the ATOMS AI Literacy Group**).
- Personnel who become aware of AI misuse or compliance violations should report through institutional channels (**IAMS Administrative Committee and/or Academic Committee**).
- Non-compliance may be subject to disciplinary measures in accordance with Academia Sinica regulations and applicable laws.
- Existing institutional compliance, audit, and reporting mechanisms apply to AI-related activities where relevant.
- Where required for compliance, transparency, or reproducibility, records of AI tool use should be maintained in accordance with institutional and external requirements.

8. References

1. Artificial Intelligence Basic Act, Taiwan (人工智慧基本法), effective January 14, 2026. https://law.nstc.gov.tw/LawContent.aspx?id=GL000592

2. Academia Sinica, "Guidelines for Using Generative AI Tools," February 2026. https://www.iias.sinica.edu.tw/storage/upload/ck_files/Guidelines_for_Using_Generative_AI_Tools.pdf

3. European Commission, "Responsible Use of Generative AI in Research: Living Guidelines," 2026. https://research-and-innovation.ec.europa.eu/document/download/2b6cf7e5-36ac-41cb-aab5-0d32050143dc_en

4. COPE Council, "COPE Position Statement on Authorship and AI Tools," 2023. https://publicationethics.org/cope-position-statements/ai-author